

## Project 10

# Branch-Dependent Planning with the PUMA Positioning Arm

**Format:** Group of two students; simulation-based project.

**Course foundations:** Lectures 3–7; Exercise 3.

## Motivation

A wrist-center position may be reached by several shoulder/elbow postures. Those alternatives can have different continuations along the same Cartesian curve. Restricting PUMA to its three-joint positioning task makes the global singularity structure visible while retaining the planning question: which initial solution can follow the complete path?

## Task

**Overall objectives.** Treat PUMA's first three joints as a noncuspidal 3R wrist-center positioning robot and explain path feasibility through reduced singularity maps and the distribution of its IK solutions.

- **Extract the position task.** Identify the wrist center from the PUMA axis geometry and derive its position as a function of  $q_1, q_2, q_3$ . Validate the D–H and URDF-based descriptions. The wrist joints are irrelevant to this point's position; do not replace the wrist center with the tool tip.
- **Build the global reference maps.** Derive  $J_p$  and plot  $\det J_p$  and its zero contours in  $(q_2, q_3)$ . Map the singular curves into  $(\rho, z)$  and count distinct arm IKs. Identify two-/four-IK regions where present and explain if this specific geometry lacks one of those regions.
- **Select and classify initial solutions.** Choose one regular loop and two translated or scaled variants approaching different arm singularities. Enumerate the initial arm IKs and identify their shoulder/elbow categories. Overlay them on the reduced plot and record mathematical versus joint-limit-admissible configurations.
- **Continue every admissible branch.** Follow each workspace curve from each valid initial arm configuration. Use warm-started continuation and analytical enumeration as a cross-check. Refine the step size near critical events; preserve branch identity instead of replacing a failed solution with an unrelated IK at the next sample.
- **Explain completion and failure.** Compare completion fraction, joint travel, joint-limit margin, and the smallest singular value of  $J_p$ . Relate each failed lift to its configuration-space separator and workspace critical curve. A workspace point can have another regular IK even when the selected continuation fails.

## Specifics and scope

Use one PUMA positioning model and three position curves, with about 200 samples per curve before local refinement. Keep orientation outside the task and use the same joint limits and tolerances for every comparison. Explore mathematical branch structure first, then report limit filtering. The goal is an explanation of branch-dependent feasibility, not a new general path-search algorithm.

Use the position Jacobian  $J_p$  for the 3R task. Confirm that the meridian reduction is valid for the selected model and operational point. Count distinct real IKs modulo periodicity, then filter joint limits separately. Identify two-/four-IK regions where present and explain any absent count rather than forcing both types into every map. Show both determinant values and the zero contour, and distinguish sampled connectivity from a proof. Refine the decisive region instead of claiming completeness from a coarse grid.

## Expected outcome

Understand the PUMA arm's D–H model, noncuspidal branch separation, and the difference between workspace reachability and a feasible lift. Deliver reduced determinant maps with overlaid IKs and trajectories, workspace IK-count maps, and a branch-by-branch path comparison.

Submit reproducible code, model parameters and test data, the required plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful cases and failures; conclusions must follow the numerical and geometric evidence rather than an expected outcome.

## Resources

The PUMA 560 URDF is `lectures_main/assets/models/puma/puma560_robot.urdf`; its STL meshes are in the same directory. The lecture website provides axis/frame inspection, IK visualization, and path demonstrations. Confirm joint offsets and the selected terminal frame when importing the model. The Lecture 7 PUMA demonstration already follows wrist-center position with fixed wrist joints and is directly relevant to this project. Reuse its continuation and visualization as a starting point, checking the analytical arm IK against Lectures 3–4.

Use the lecture website to verify D–H parameters, visualize IKs and singularities, and simulate paths. All listed paths are relative to the repository root. Start the website following `lectures_main/README.md`.

## Workload and collaboration

Budget approximately 60 hours per student (120 person-hours per pair) for this 2-ECTS project, following EPFL's 30-hours-per-credit convention.<sup>1</sup> Deduct any lectures or exercises included in those credits. Allocate roughly 20 team-hours to model validation, 50 to the core work, 30 to experiments, and 20 to reporting. Reuse the specified IK and simulations. Share validation and interpretation even if modelling and implementation are divided between students. Hardware execution is outside scope.

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<sup>1</sup>EPFL: Design a Course, workload guidance.